

AI-Powered Customer Support Ticketing System

Salesforce-Based Customer Support Automation Solution

1. Project Overview

The AI-Powered Customer Support Ticketing System is a Salesforce-based application designed to streamline customer support operations. The system manages customer issues through Case Management, automation workflows, SLA tracking, and AI-powered agent assistance.

The solution uses Salesforce standard objects, custom objects, record-triggered flows, security configurations, reports, and Agentforce AI concepts to improve support efficiency and reduce manual effort.

2. Problem Statement

Customer support teams often face challenges such as manual ticket classification, delayed responses, SLA violations, and difficulty tracking agent performance. A centralized automated system was required to organize customer requests, prioritize issues, support agents, and improve resolution speed.

3. Requirement Gathering

Key requirements identified:

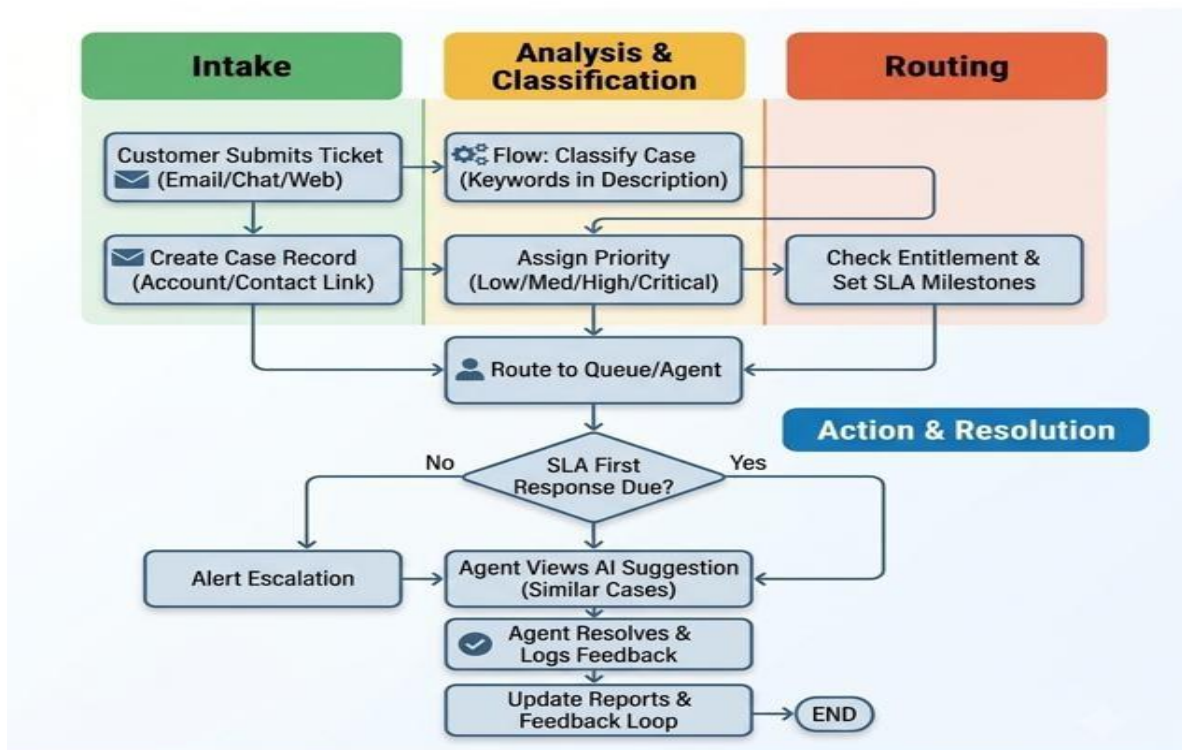
- ✓ Centralized customer support case management.
- ✓ Automatic case classification based on issue details.
- ✓ Priority assignment and SLA deadline tracking.
- ✓ Agent assistance through AI-based suggestions.
- ✓ Role-based access control.
- ✓ Reports for monitoring support performance.
- ✓ Feedback collection for continuous improvement.

4. System Architecture

The architecture represents the complete support lifecycle from customer ticket creation to resolution and improvement.

Architecture Flow:

Customer Request → Salesforce Case Creation → Automation & Classification → SLA Management → Agentforce AI Assistance → Agent Resolution → Reports → AI Feedback Loop.



5. Functional Scope

The system provides:

- ✓ Customer case creation and tracking.
- ✓ Automatic classification and priority assignment.
- ✓ SLA due date calculation.
- ✓ Agent suggested actions.
- ✓ Case resolution management.
- ✓ Feedback capture.
- ✓ Reports and dashboards.
- ✓ Role-based security management.

6. Stakeholder Mapping

- Customer: Creates support requests and receives resolutions.
- Support Agents: Handle assigned cases, use AI suggestions, update resolutions, and provide feedback.
- Support Managers: Monitor SLA compliance, agent performance, and reports.
- System Administrators: Configure objects, flows, security, users, and system maintenance.
- Business Stakeholders: Define requirements and validate business objectives.

7. Execution Roadmap

- Phase 1: Requirement Analysis & Planning
Identify business needs, scope, stakeholders, and architecture.
- Phase 2: Salesforce Environment Setup
Create Developer Org and configure Salesforce settings.
- Phase 3: Data Model & Security Configuration
Create objects, fields, relationships, profiles, and permission sets.
- Phase 4: Automation Implementation
Develop flows for classification, priority assignment, and SLA tracking.
- Phase 5: Agent Assistance & Feedback Loop
Configure AI suggestions and feedback capture.
- Phase 6: Reporting & Monitoring
Create dashboards and reports.
- Phase 7: Testing & Validation
Perform functional and automation testing.
- Phase 8: Deployment & Documentation
Finalize configuration and prepare demonstration.

8. System Requirements

Software:

- ✓ Salesforce Developer Org
- ✓ Lightning Experience
- ✓ Salesforce Flow Builder
- ✓ Reports and Dashboards

Functional Requirements:

- ✓ Case management
- ✓ SLA tracking
- ✓ Automation
- ✓ AI assistance
- ✓ Security controls

Non-Functional Requirements:

- ✓ Scalability
- ✓ Data accuracy
- ✓ Reliability

- ✓ Easy maintenance

9. Agentforce AI Integration

Agentforce AI capabilities support agents by providing intelligent assistance during case resolution.

Implemented AI-style features:

- ✓ Suggested responses/actions.
- ✓ Similar case guidance.
- ✓ Feedback collection from agents.
- ✓ Continuous improvement through feedback analysis.

10. Data Architecture & Object Model

Standard Objects:

- ✓ Account - Stores customer organization information.
- ✓ Contact - Stores customer details.
- ✓ Case - Stores support requests.

Custom Objects:

- ✓ Support_Category__c - Maintains issue categories.
- ✓ SLA_Policy__c - Defines SLA rules.
- ✓ AI_Feedback__c - Stores agent feedback.

Relationships:

- ✓ Account → Contact → Case
- ✓ Support Category → Case
- ✓ SLA Policy → Case
- ✓ AI Feedback → Case

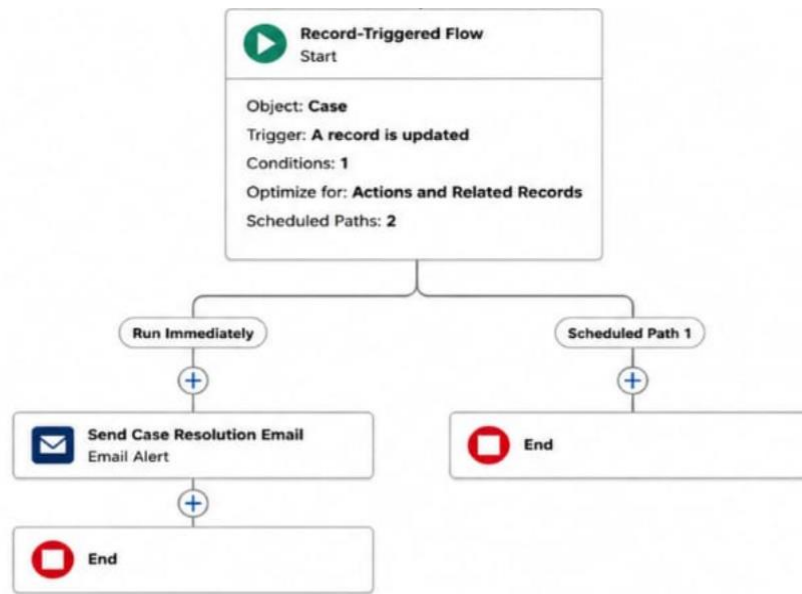


Figure 3: Record-Triggered Flow for Case Resolution Email

This record-triggered flow automatically sends a case resolution email when a case record is updated. The automation ensures timely communication and improves customer satisfaction.

11. Automation Logic

Record-triggered flows automate:

- ✓ Case classification using issue description.
- ✓ Priority assignment.
- ✓ SLA due date calculation.
- ✓ Case routing to queues or agents.
- ✓ Resolution email notification.
- ✓ Feedback capture workflow.

12. Security Model

Security is implemented using Salesforce profiles and permission sets.

Profiles:

- ✓ Support Agent
- ✓ Support Manager
- ✓ System Administrator

Permission Sets:

- ✓ AI Feature Access
- ✓ Report Access

Security controls include object permissions, field-level security, and role-based access.

13. Report and Monitoring

Reports and dashboards provide visibility into:

- ✓ Open and closed cases.
- ✓ SLA compliance.
- ✓ Priority-based case distribution.
- ✓ Agent performance.
- ✓ Resolution trends.

14. AI Feedback Loop

Agent feedback is captured after using AI suggestions. This feedback helps evaluate suggestion quality and improves future support processes.

15. Testing

Testing includes:

- ✓ Case creation testing.
- ✓ Validation rule testing.
- ✓ SLA calculation testing.
- ✓ Flow execution testing.
- ✓ Email notification testing.
- ✓ Security and permission testing.

16. Maintenance and Troubleshooting

Maintenance activities:

- ✓ Review flows and automation.
- ✓ Update fields and configurations.
- ✓ Audit security access.
- ✓ Monitor reports.

Troubleshooting:

- ✓ Analyze flow errors.
- ✓ Review debug logs.
- ✓ Check validation rules.
- ✓ Correct data configuration issues.

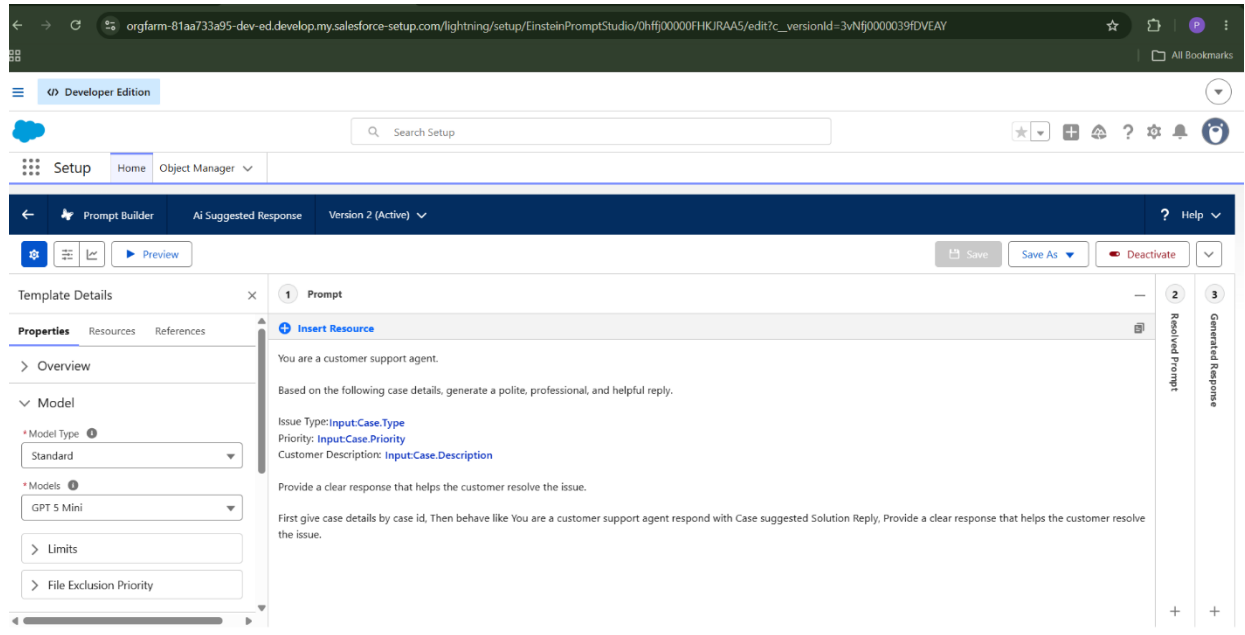
17. Learning Outcomes

- ✓ Practical Salesforce Administration experience.
- ✓ Understanding of data modeling.
- ✓ Hands-on Flow automation.
- ✓ Security configuration knowledge.
- ✓ Experience designing AI-assisted business solutions.

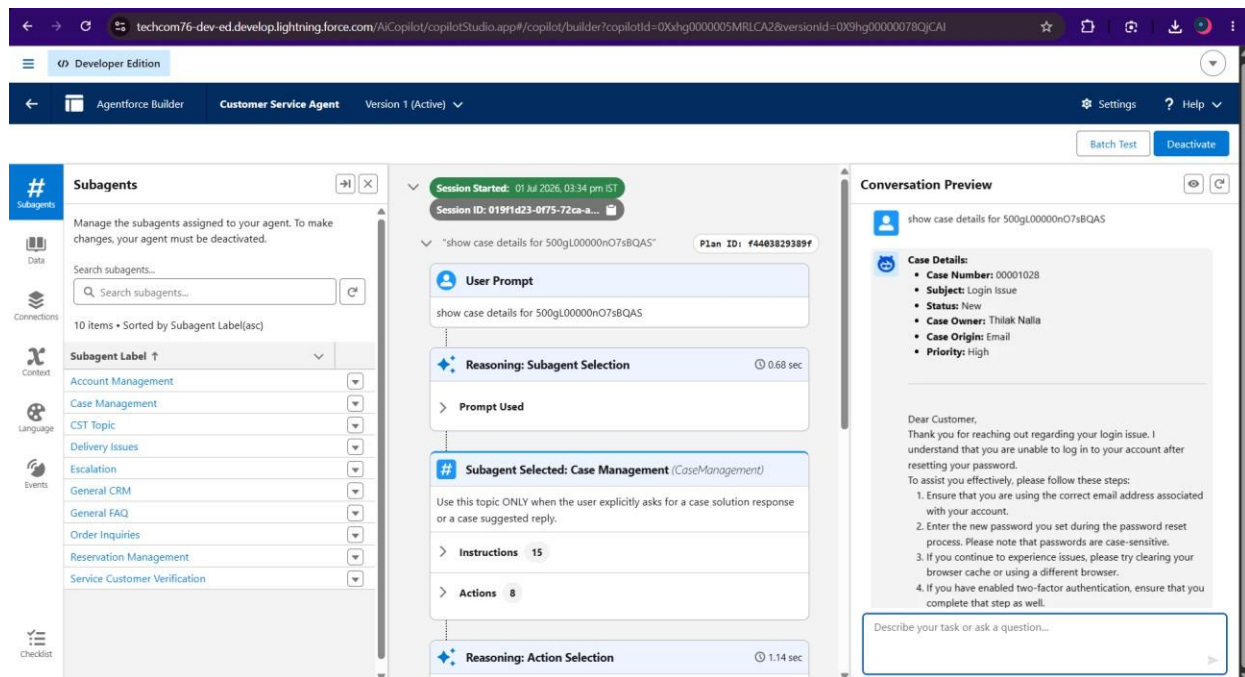
18. Output of the Project

The final system delivers an automated customer support platform that reduces manual ticket handling, improves SLA management, supports agents with intelligent suggestions, and provides management visibility through analytics.

Model Used:



Output:



19. Conclusion

The AI-Powered Customer Support Ticketing System successfully demonstrates how Salesforce automation and AI-assisted support capabilities can improve customer service operations. The solution provides structured case management, faster resolution, SLA tracking, secure access control, and continuous improvement through feedback mechanisms.